UEI745 Deep Learning in Health Care Applications

L T P Cr

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Course Objective: This course aims to not only cover the fundamentals of deep learning, butalso give a grasp of contemporary research related to healthcare.

Introduction: Overview of machine learning and deep learning, linear classifiers, lossfunctions, Importance of machine and deep learning techniques in medical applications **Optimization:** Stochastic gradient descent and contemporary variants, back-propagation.

Feedforward networks and training: Activation functions, initialization, regularization,batch normalization, model selection, ensembles.

Convolutional neural networks: Fundamentals, architectures, pooling, visualization,Arrythmia detection using CNN

Recurrent neural networks: Recurrent neural networks (RNN), long-short term memory (LSTM), Deep learning for Health Informatics, Application in Medical image and signal classification

Transfer Learning: Overview of transfer learning**,** Application of transfer learning in medicalimage recognition, signal classification

Deep Learning in Medical Application: Differential diagnosis, Learning a Health KnowledgeGraph from Electronic Medical Records Machine learning for cardiology: Fully Automated Echocardiogram Interpretation in Clinical Practice

Course Learning Outcomes (CLO): On successful completion of this module, the studentshould be able to:

1. Analyze and evaluate the concepts of machine learning
2. Develop convolutional neural network for the medical image analysis
3. Develop recurrent neural network for medical images and signal classification
4. Develop different transfer learning models for healthcare application

Text Books:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville (2016). *Deep Learning,* MIT Press
2. K. P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
3. C. M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006

Evaluation Scheme

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Sr..No	Evaluation Elements	Weightage (%)
1	MST+EST	70
3	Sessional (Lab, Quiz,Assignment)	30

List of Experiments

1. Implement Support Vector Machine classifier for signal classification
2. Implementation of Neural Network in Medical Image classification.
3. Design CNN model for Arrythmia detection
4. Implementation of RNN in medical signal classification
5. Implementation of LSTM model in medical image classification
6. Application of transfer learning in seizure detection
7. Implementation of CNN-LSTM model in medical image captioning
8. Implementation of Pre-trained model (VGG16, Alexnet, Googlenet) forfeature extraction from signal
9. Application of VGG16 in EMG signal classification
10. Compare the accuracy achieved by deep learning model and transfer learningmodel in seizure detection